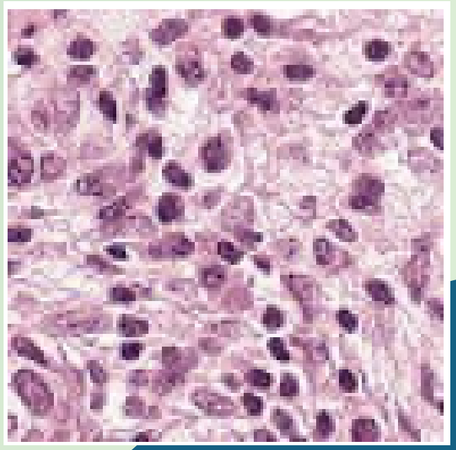
Topic : Histopathologic Cancer Detection

Objective:
Develop a deep learning model using Convolutional Neural Networks (CNNs) to classify histopathologic images of lymph node sections as either:

- Containing metastatic cancer (label 1)
- Not containing metastatic cancer (label 0)



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- **Significance:**
- **Early Detection**: Effective treatment planning depends on timely and accurate detection of cancer.
- **Clinical Impact**: A reliable model supports pathologists in quickly identifying metastasis, facilitating prompt and appropriate interventions.
- **Automation Potential**: Automating classification could reduce manual analysis time and costs in pathology, enhancing lab efficiency.

Approach:

- **Binary Image Classification Task**: This project is structured as a binary classification problem, similar to standard image classification tasks like CIFAR-10 and MNIST.
- **Model Design**: A Convolutional Neural Network (CNN) is used for feature extraction and classification, differentiating metastatic from non-metastatic lymph node images.

Procedure :

Phase 1:Data Exploration

Phase 2: Model Making

Phase 3 : Model Evaluation

Dataset Overview

Challenges in the Dataset:

- **Large Dataset**: Approximately 200,000 images, requiring significant computational resources for processiang.
- **High Dimensionality**: Images are 96x96x3, which is more complex than smaller standard datasets like MNIST (28x28), increasing computational and memory demands.

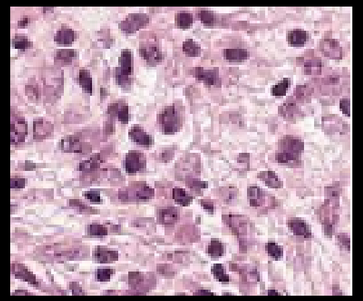
Description:

- **Size**: Approximately 220,000 images
- **Resolution**: 96x96 pixels
- **Classes/Labels**: Binary labels indicating:
  - **Label 1**: Metastatic (cancer present)
  - **Label 0**: Non-metastatic (no cancer detected)

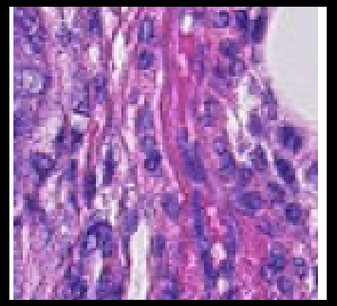
Physical Visual Observations:

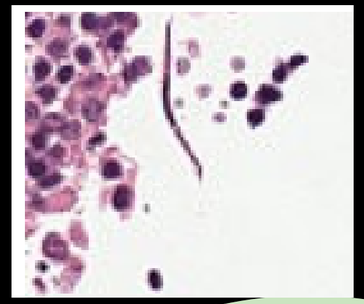
- **Non-Cancerous Cells**: Tend to have a more regular, defined shape.
- **Cancerous Cells**: Often exhibit irregular, undefined shapes, which may aid in distinguishing between the two classes during model training.

Non-Cancerous



Cancerous Cells





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TOTAL IMAGES	LABEL DISTRIBUTION
276025	Label 0: 130908, Label 1: 89117

Model Making

Data Augmentation with ImageDataGenerator

Purpose of Data Augmentation:

- Data augmentation helps increase the effective size and variability of the training dataset.
- It prevents overfitting by exposing the model to different variations of the images.

Data Augmentation Techniques

- Rescaling: All pixel values are scaled to the range [0, 1] for normalization, which aids in stable training.
- Horizontal Flip: Randomly flips images horizontally, providing left-right variations of the same image.
- Vertical Flip: Randomly flips images vertically, introducing top-down variations.
  - Applies rescaling, horizontal flipping, and vertical flipping to images.
  - Ensures the model sees a variety of transformations, increasing robustness.
- **Validation Data Generator (val_datagen)**:
  - Only rescaling is applied, ensuring consistent evaluation without data distortion.

Flow from Dataframe

- **train_generator**: Generates batches of training images from the train_labels dataframe.
  - **Parameters**:
    - target_size: (96, 96) - Resizes images to a standard size.
    - batch_size: 32 - Specifies the number of images in each batch.
    - class_mode: 'binary' - Suitable for two-class (binary) classification.
- **val_generator**: Similar setup for validation images to

Baseline vs. Advanced Model Architecture

Baseline Model Architecture

Purpose

- Basic CNN for binary classification.

Architecture

- **Input**: 96x96 RGB images.
- **Conv Layers**:
  - 1st: 32 filters, 3x3, ReLU
  - 2nd: 64 filters, 3x3, ReLU
- **Pooling**: MaxPooling (2x2) after each Conv layer.
- **Regularization**: No Dropout (simpler but may risk overfitting).
- **Dense Layer**: 128 units, ReLU.
- **Output Layer**: 1 unit, Sigmoid (0 or 1).

Advanced Model Architecture

Purpose

- Enhanced CNN to improve performance and reduce overfitting.

Architecture

- **Input**: 96x96 RGB images.
- **Conv Layers**:
  - 1st: 32 filters, 3x3, ReLU
  - 2nd: 64 filters, 3x3, ReLU
  - 3rd: 128 filters, 3x3, ReLU (captures complex patterns)
- **Pooling**: MaxPooling (2x2) after each Conv layer.
- **Regularization**:
  - 20% Dropout after each Conv layer
  - 50% Dropout before final dense layer
- **Dense Layer**: 256 units, ReLU.
- **Output Layer**: 1 unit, Sigmoid (0 or 1).

Baseline Model Results

Final Loss: 0.3080
Final Validation Loss: 0.3129
Observations:

- Initial training accuracy started low but improved steadily.
- Achieved good accuracy but showed signs of moderate overfitting with validation loss slightly higher than training loss.

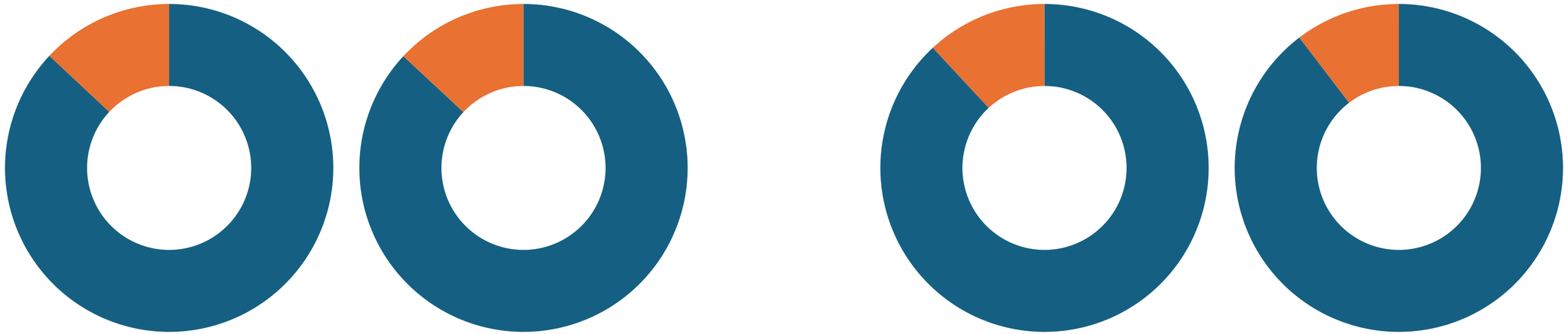
Advanced Model Results

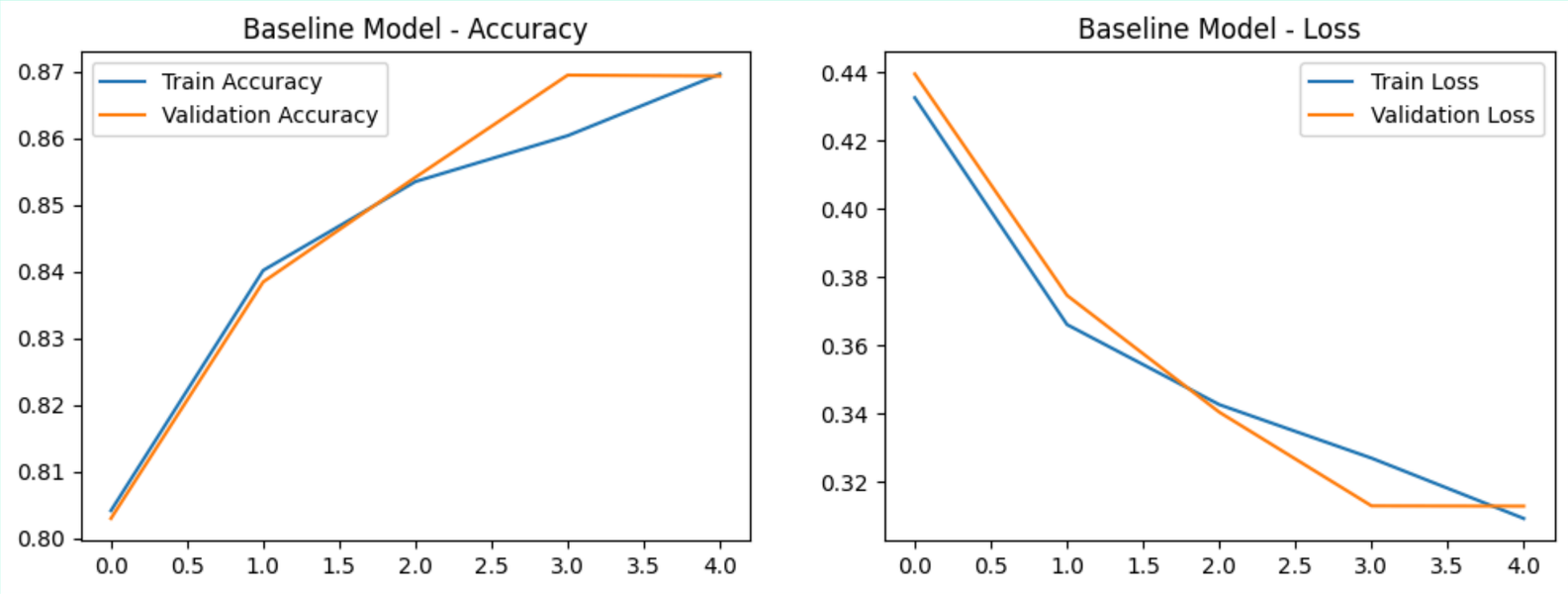
Final Loss: 0.2940
Final Validation Loss: 0.2565
Observations:

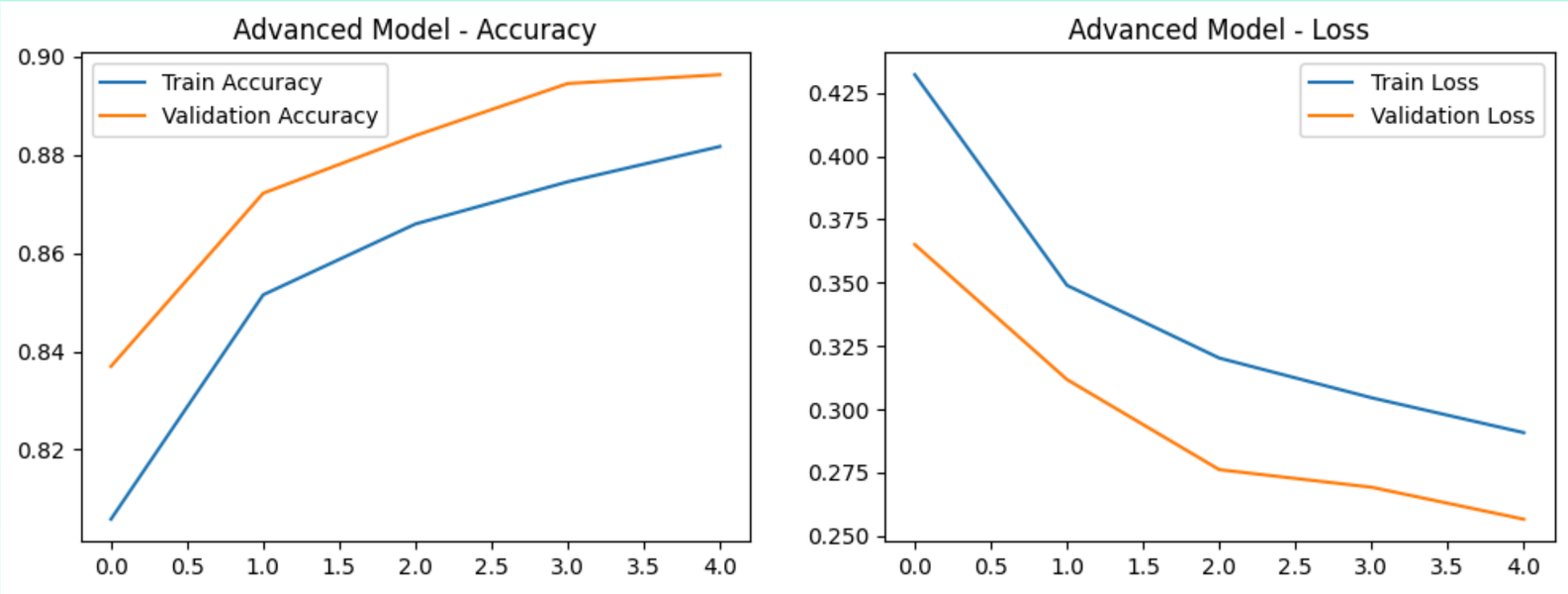
- Higher accuracy and lower loss compared to the baseline model, both in training and validation.
- Advanced model's regularization (dropout) reduced overfitting, as seen in the smaller gap between training and validation metrics.

Overall Observations:

The advanced model demonstrated significantly superior performance compared to the baseline model across both training and validation sets. It achieved higher accuracy and lower loss values, indicating a more effective learning process. The introduction of dropout layers played a crucial role in reducing overfitting, as evidenced by the improved balance between training and validation performance. This architectural change allowed the model to generalize better to unseen data while still maintaining accuracy on the training dataset.



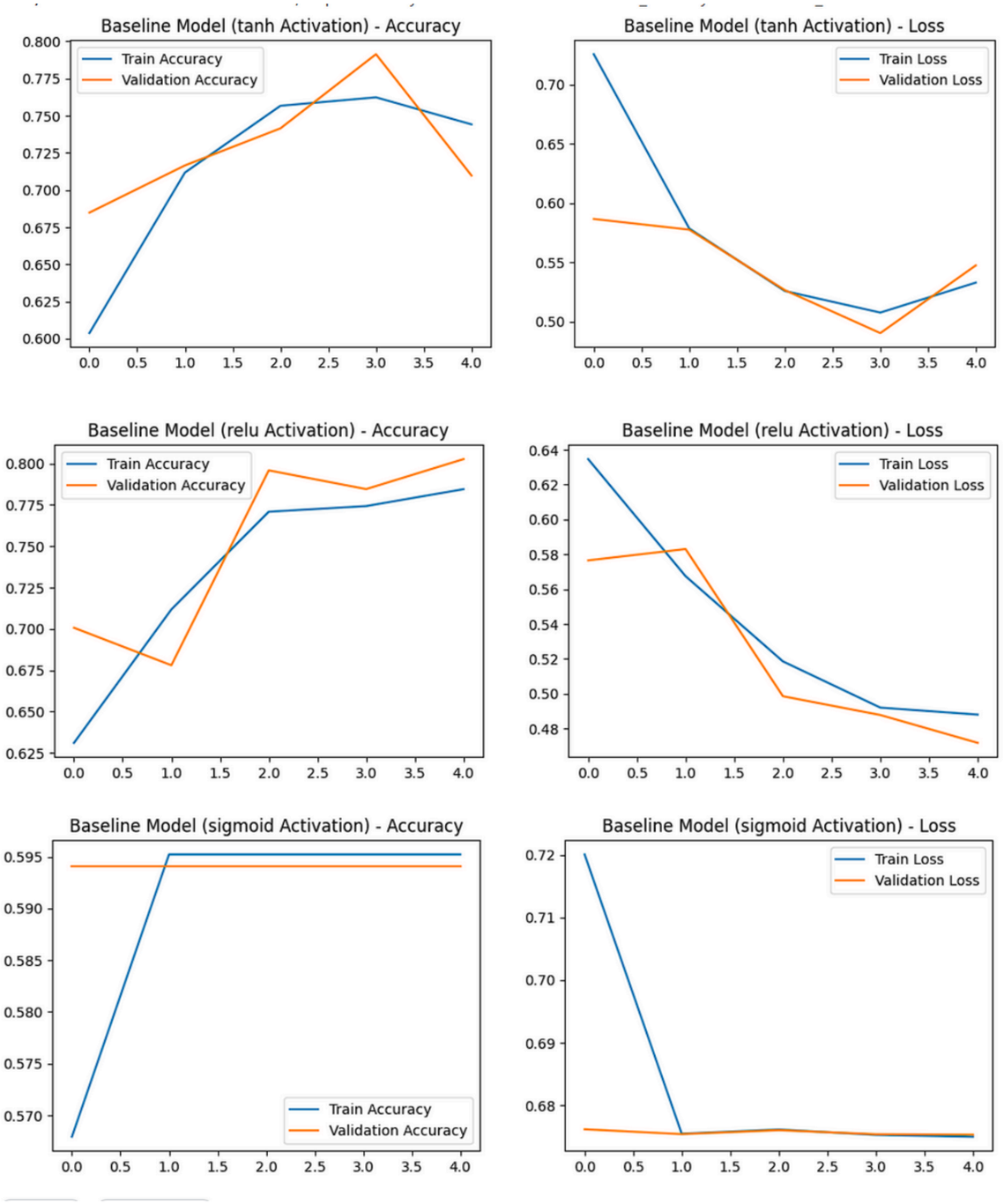

Baseline Model - Accuracy
Train Accuracy
Validation Accuracy
0.87
0.86
0.85
0.84
0.83
0.82
0.81
0.80
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Baseline Model - Loss
Train Loss
Validation Loss
0.44
0.42
0.40
0.38
0.36
0.34
0.32
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0


Advanced Model - Accuracy
Train Accuracy
Validation Accuracy
0.90
0.88
0.86
0.84
0.82
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Advanced Model - Loss
Train Loss
Validation Loss
0.425
0.400
0.375
0.350
0.325
0.300
0.275
0.250
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0

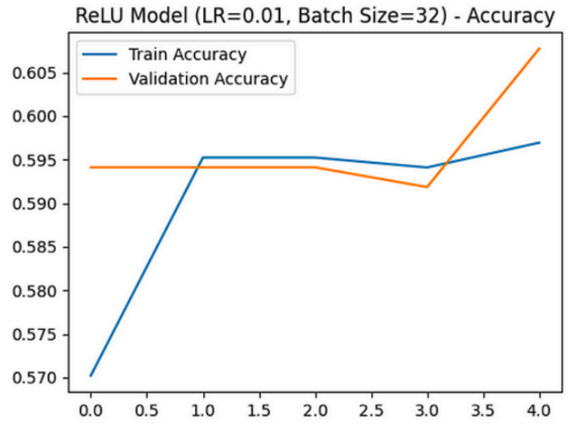
HYPER PARAMETER TUNNING

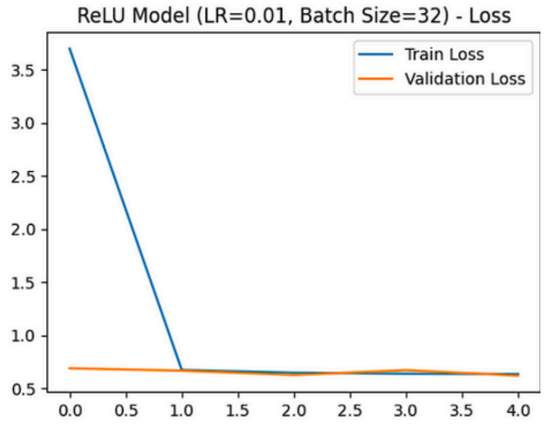
BASELINE MODEL TRAINED FOR DIFFERENT ACTIVATION FUNCTION
tanh , Relu and Sigmoid

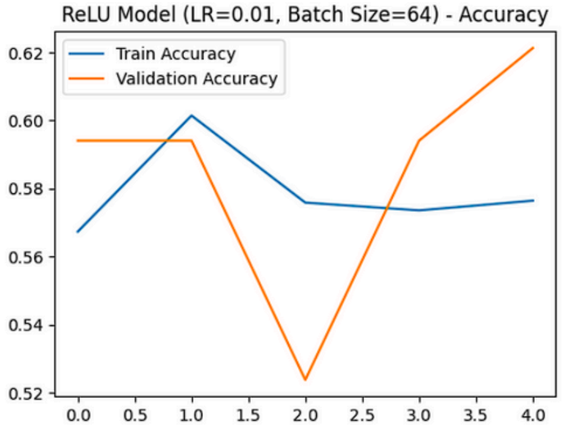
Found that Relu has a maximum accuracy

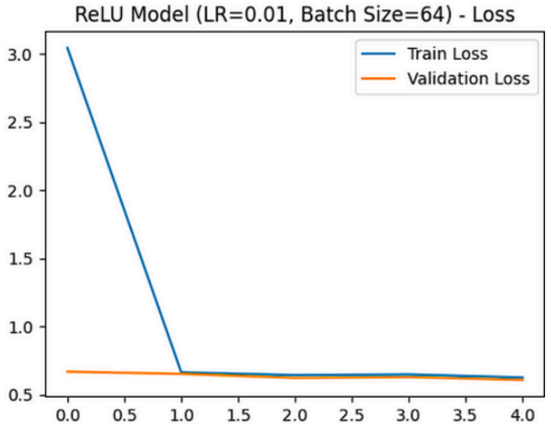


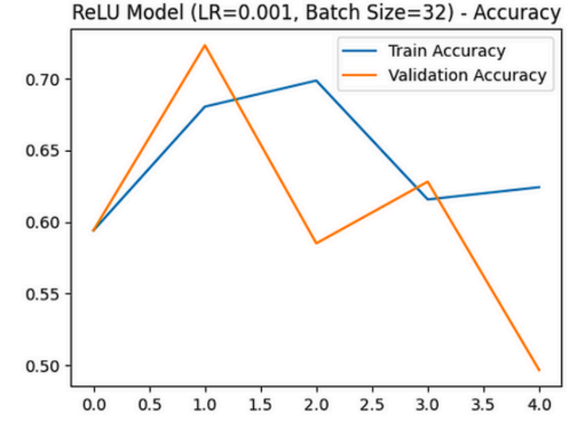
***TRAINED FOR 0.01% DATA ONLY**

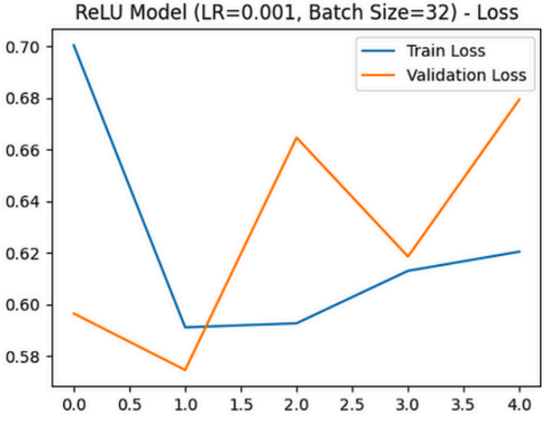

ReLU Model (LR=0.01, Batch Size=32) - Accuracy
Train Accuracy
Validation Accuracy

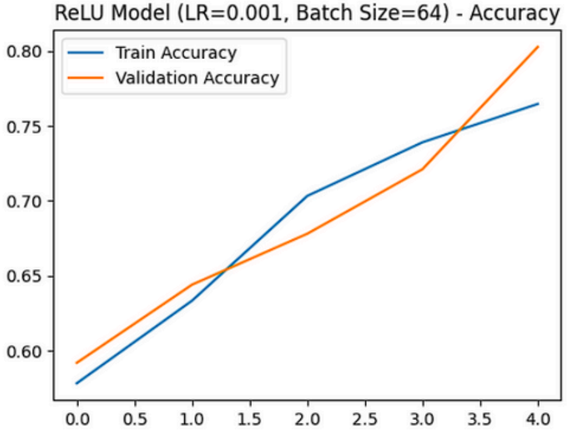

ReLU Model (LR=0.01, Batch Size=32) - Loss
Train Loss
Validation Loss

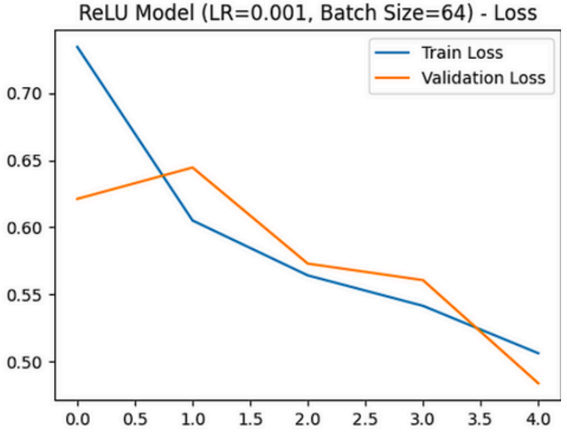

ReLU Model (LR=0.01, Batch Size=64) - Accuracy
Train Accuracy
Validation Accuracy

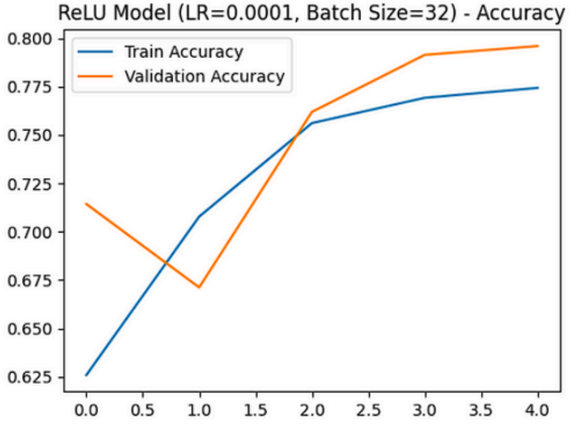

ReLU Model (LR=0.01, Batch Size=64) - Loss
Train Loss
Validation Loss


ReLU Model (LR=0.001, Batch Size=32) - Accuracy
Train Accuracy
Validation Accuracy


ReLU Model (LR=0.001, Batch Size=32) - Loss
Train Loss
Validation Loss


ReLU Model (LR=0.001, Batch Size=64) - Accuracy
Train Accuracy
Validation Accuracy


ReLU Model (LR=0.001, Batch Size=64) - Loss
Train Loss
Validation Loss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Train Accuracy
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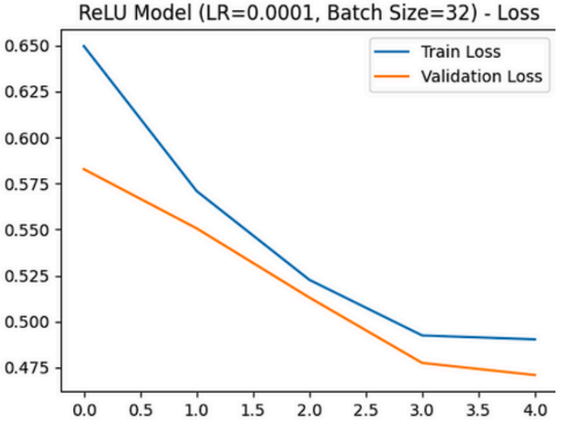
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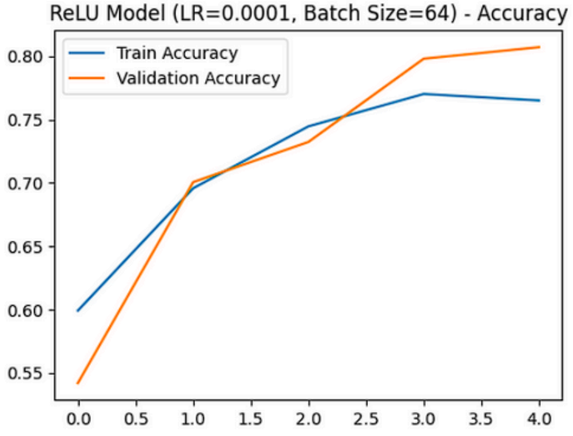
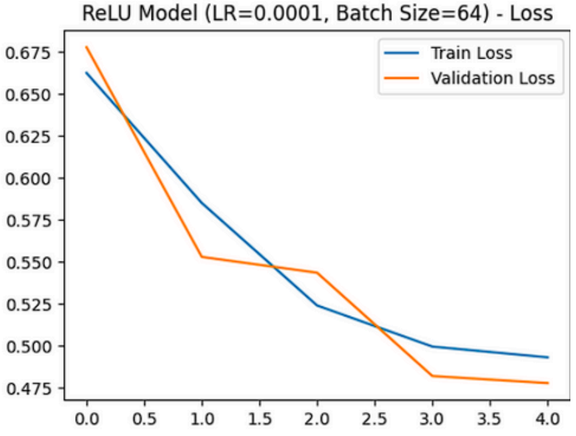
Train Loss
Validation Loss


ReLU Model (LR=0.0001, Batch Size=64) - Accuracy
Train Accuracy
Validation Accuracy


ReLU Model (LR=0.0001, Batch Size=64) - Loss
Train Loss
Validation Loss